Class Meeting Handout #10

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Ester hydrolysis can occur via a base-catalyzed mechanism, an acid-catalyzed mechanism and other possible uncatalyzed mechanisms. the rates of each of the catalyzed mechanisms will change with increasing basicity or acidity. We can plot rate vs. acidity and use this information to interpret the mechanism.

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pH-Rate Profiles

Today we will explore the log-log plot using the example of pH-rate profiles.[1] pH is a log scale - every integer step is a 10-fold change in acidity. So if we take the log of the rate or the rate constant then we can match this and a slope of one indicates a first-order dependance on acid or base. Often we can see different mechanisms dominating the overall rate at different pH values.

[1] Read chapter 9 of the textbook. Some examples are presented in figures 9.7, 9.9 and 9.11 of the textbook. For an example of the pH-rate profile for imine hydrolysis see chapter 10.15.2. We will be exploring imine chemistry in more detail later in this course. Ester hydrolysis is discussed in chapter 10.17 of the textbook and we will also be revisiting this subject later in this course.

Exercise: The Pieces – 30 minutes

Let us plot the rate vs. pH for rate laws that we established for ester hydrolysis. Recall what we discussed in the previous class meeting where we proposed mechanisms for hydrolysis of an ester and proposed rate laws for the various forms of catalysis.

1. Consider the specific acid catalysis of hydrolysis of phenyl acetate.

   (a) Write out the mechanism[2] that corresponds to scheme 1 and the rate law presented in equation 1.

$$\mathrm{E} + \mathrm{H^+} \underset{\mathrm{K_{a_{EH^+}}}}{\rightleftharpoons} \mathrm{EH^+}$$

$$\mathrm{EH^+} \xrightarrow{\mathrm{k_{H^+}}} \mathrm{P}$$

Scheme 1: Specific acid catalysis for hydrolysis of an ester

$$-\frac{d[\mathrm{E}]}{dt} = k_{EH^+}\frac{a_{H^+}}{K_{a_{EH^+}} + a_{H^+}}[\mathrm{E}]$$

Equation 1: Specific acid catalysis for hydrolysis of an ester

   (b) Apply the values in table 1 and plot the pH-rate profile for this reaction in the range $0 < p\mathrm{H} < 10$.

[2] Whenever I ask for a mechanism, I mean a step-by-step mechanism including curved arrows to show electron movement.

Table 1: Kinetic parameters for hydrolysis of phenyl acetate.

$K_{a_{EH^+}}$	3.2×10^{6} M
k_{EH^+}	$6510\,\mathrm{s^{-1}}$
k_E	$1.2 \times 10^{-5}\,\mathrm{s^{-1}}$
k_{OH^-}	$9.3\,\mathrm{s^{-1}M^{-1}}$
$K_{a_{AcOH}}$	3.2×10^{-5} M
k_{AcO^-}	$5.6 \times 10^{-3}\,\mathrm{s^{-1}M^{-1}}$
[AcOH]	0.10 M

2. Consider the specific base catalysis of hydrolysis of phenyl acetate.

   (a) Write out the mechanism that corresponds to scheme 2 and the rate law presented in equation 2.

$$\mathrm{E} + \mathrm{OH}^- \xrightarrow{\mathrm{k_{OH^-}}} \mathrm{P}$$

Scheme 2: Specific base catalysis for hydrolysis of an ester

$$-\frac{d[\mathrm{E}]}{dt} = k_{OH^-}[\mathrm{OH}^-][\mathrm{E}]$$

Equation 2: Specific base catalysis for hydrolysis of an ester

   (b) Apply the values in table 1 and plot the pH-rate profile for this reaction in the range $0 < p\mathrm{H} < 10$.

3. Hydrolysis of esters is very slow in neutral water, but phenyl acetate is a very reactive ester. Will the water rate be significant enough to be observable? Consider the uncatalyzed hydrolysis route.

   (a) Write out the mechanism that corresponds to scheme 3 and the rate law presented in equation 3.

$$\mathrm{E} \xrightarrow{\mathrm{k_E}} \mathrm{P}$$

Scheme 3: Uncatalyzed for hydrolysis of an ester

$$-\frac{d[\mathrm{E}]}{dt} = k_E[\mathrm{E}]$$

Equation 3: Uncatalyzed hydrolysis of an ester

   (b) Apply the values in table 1 and plot the pH-rate profile for this reaction in the range $0 < p\mathrm{H} < 10$.

4. When hydroxide concentrations are small we may still see general-base catalyzed hydrolysis by buffer. Imagine that we used acetate buffer in this experiment. Consider a general base catalyzed hydrolysis of an ester.

   (a) Write out the mechanism that corresponds to scheme 3 and the rate law presented in equation 3.

$$\mathrm{AcOH} \xrightleftharpoons{\mathrm{K_{a_{AcOH}}}} \mathrm{AcO}^- + \mathrm{H}^+$$
$$\mathrm{E} + \mathrm{AcO}^- \xrightarrow{\mathrm{k_{AcO^-}}} \mathrm{P}$$

Scheme 4: General base catalysis by acetate anion for hydrolysis of an ester

$$-\frac{d[\mathrm{E}]}{dt} = k_{AcO^-}\frac{K_{a_{AcOH}}}{K_{a_{AcOH}} + a_{H^+}}[\mathrm{AcOH}][\mathrm{E}]$$

Equation 4: General base catalysis by acetate anion for hydrolysis of an ester

   (b) Apply the values in table 1 and plot the pH-rate profile for this reaction in the range $0 < p\mathrm{H} < 10$.

Exercise: Add It Up – 20 minutes

All of these reactions are occurring simultaneously.

5. Produce a pH-rate profile that is the sum of all four reactions.[3]
6. Change the rate constants and acid dissociation constants and see how the pH-rate profile changes. For example reduce the value of k_{AcO^-} or k_E 1000-fold. Fool around and find out what happens.
7. Every possible reaction that you can imagine is happening in any system. But usually only one occurs at a rate that matters. How does that statement fit with what you have learned about hydrolysis reactions in water at various pH values?

[3] See the moodle site for the *Python* notebook used during this class meeting. I hope you will see how easy it is to use a tool like *Python* for data visualization and analysis. You don't need to be a programmer, but you will need to become a user of computational tools as you transition from being a learner of chemistry to a doer of chemistry.

Learning Goals

There is more to a class meeting than what we do in 50 minutes of class time. After participating in the above exercises and completing the requirements described on the moodle site you will have achieved the following...

1. Understand the difference between specific and general acid/base catalysis.
2. Construct rate laws for specific and general acid/base catalysis and be able to interpret reaction kinetics to confirm the rate law.
3. Be able to interpret a pH-rate profile.

Looking Ahead

General and specific catalysis is a common feature in many reactions that occur in water, especially enzyme-catalyzed reactions. This topic will be essential in interpreting enzyme mechanisms and kinetics. We can also investigate the effect of changing the base itself. We can plot the $p\mathrm{K}_a$ of the various bases used against the log of the rate. From this we can interpret whether proton transfer os complete in the transitions state (specific catalysis), partial (general catalysis) or not important (no acid/base catalysis). This is the famous Brønsted plot and this method will also serve as our first example of a linear free-energy relationship (LFER), which will be the focus of the next two weeks of this course.

Exercise: A Final Challenge

We won't have time to do this during the class meeting and this is not a high school math class, but try reviewing your basic math skills by completing the following exercise.[4]

The final rate law could be written as...

[4] It's never calculus that trips up students – it's algebra.

Equation 5: The total rate law for all pathways combined.

$$-\frac{d[\mathrm{E}]}{dt} = k_{obs}[\mathrm{E}]$$

$$k_{obs} = k_{EH^+}\frac{a_{H^+}}{K_{a_{EH^+}} + a_{H^+}} + k_{OH^-}[\mathrm{OH^-}] + k_E + k_{AcO^-}\frac{K_{a_{AcOH}}}{K_{a_{AcOH}} + a_{H^+}}[\mathrm{AcOH}]$$

...or by using the powers of algebra, we can obtain the following polynomial version of the rate law. Is it equivalent? Check my work by doing the math.

Equation 6: The total rate law rearranged into polynomial form.

$$\begin{aligned} k_{obs} = {} & \frac{\left(k_{EH^+} + k_E\right) a_{H^+}^3}{a_{H^+}^3 + \left(K_{a_{AcOH}} + K_{a_{EH^+}}\right) a_{H^+}^2 + K_{a_{EH^+}} K_{a_{AcOH}} a_{H^+}} + \\ & \frac{\left(k_{OH^-} K_W + K_{a_{AcOH}}\left(k_E + k_{EH^+} + k_{AcO^-}[\mathrm{AcOH}]\right) + k_E K_{a_{EH^+}}\right) a_{H^+}^2}{a_{H^+}^3 + \left(K_{a_{AcOH}} + K_{a_{EH^+}}\right) a_{H^+}^2 + K_{a_{EH^+}} K_{a_{AcOH}} a_{H^+}} + \\ & \frac{\left(k_{OH^-} K_W \left(K_{a_{AcOH}} + K_{a_{EH^+}}\right) + K_{a_{EH^+}} K_{a_{AcOH}}\left(k_E + k_{AcO^-}[\mathrm{AcOH}]\right)\right) a_{H^+}}{a_{H^+}^3 + \left(K_{a_{AcOH}} + K_{a_{EH^+}}\right) a_{H^+}^2 + K_{a_{EH^+}} K_{a_{AcOH}} a_{H^+}} + \\ & \frac{K_{a_{EH^+}} K_{a_{AcOH}} k_{OH^-} K_W}{a_{H^+}^3 + \left(K_{a_{AcOH}} + K_{a_{EH^+}}\right) a_{H^+}^2 + K_{a_{EH^+}} K_{a_{AcOH}} a_{H^+}} \end{aligned}$$

It looks crazy, but it is just a 3rd-order polynomial divided by another 3rd-order polynomial and could be expressed as shown in equation 7. Which version do you believe is better for interpreting the pH-rate profile, equation 5 or equation 7?

Equation 7: The total rate law for all pathways combined.

$$k_{obs} = \frac{A \cdot a_{H^+}^3 + B \cdot a_{H^+}^2 + C \cdot a_{H^+} + D}{a_{H^+}^3 + X \cdot a_{H^+}^2 + Y \cdot a_{H^+}}$$

$$\begin{aligned} A &= k_{EH^+} + k_E \\ B &= k_{OH^-} K_W + K_{a_{AcOH}}\left(k_E + k_{EH^+} + k_{AcO^-}[\mathrm{AcOH}]\right) + k_E K_{a_{EH^+}} \\ C &= k_{OH^-} K_W \left(K_{a_{AcOH}} + K_{a_{EH^+}}\right) + K_{a_{EH^+}} K_{a_{AcOH}}\left(k_E + k_{AcO^-}[\mathrm{AcOH}]\right) \\ D &= K_{a_{EH^+}} K_{a_{AcOH}} k_{OH^-} K_W \\ X &= K_{a_{AcOH}} + K_{a_{EH^+}} \\ Y &= K_{a_{EH^+}} K_{a_{AcOH}} \end{aligned}$$

All this looked so easy when we were plotting using *Python* tools in the class meeting. *Python* makes everything easy. Someone else wrote a tool like *MatPlotLib* or *NumPy* and we get to use it. No programming – just commands.

Appendix: Python Code

Below is the *Python* code for expressing the pH-rate profile that we presented during this class meeting. I will go through it during the meeting. The interactive *Python* notebook will be available via a link on the course moodle page for this meeting.]

```
from matplotlib import pyplot as plt
import numpy as np

# Kinetic parameters
c_AcOH = 0.1        # Concentration of AcOH in molar
k_OAc = 5.6E-3      # per molar per second
Ka_HOAc = 3.2E-5    # molar
k_OH = 9.3          # per molar per second
k_E = 1.8E-5        # per second
k_EH = 6510         # per second
Ka_EH = 3.2E6       # molar
KW = 1E-14          # molar^2

x = np.linspace(0, 12, 100) # x is pH from 0 to 12 in 100 steps
H = 10**-x
OH = KW / H

# Observed rate constants for the four possible contributing mechanisms
k_obs_EH = k_EH * H / (H + Ka_EH)
k_obs_E = k_E + 0*x  # The 0*x will produce data points to match x
k_obs_OH = k_OH * OH
k_obs_OAc = k_OAc * Ka_HOAc / (H + Ka_HOAc) * c_AcOH

# Plot contributing rate constants separately
if 1:   # use "0" or "False" to skip; "1" or "True" to execute
    plt.plot(x, np.log10(k_obs_OH), "r-", linewidth = 0.5)
    plt.plot(x, np.log10(k_obs_E), "r-", linewidth = 0.5)
    plt.plot(x, np.log10(k_obs_EH), "r-", linewidth = 0.5)
    plt.plot(x, np.log10(k_obs_OAc), "r-", linewidth = 0.5)

# Plot total rate constant
# Use "1" or "0" to include or ignore compnents
total = 1*k_obs_OH + 1*k_obs_E + 1*k_obs_EH + 1*k_obs_OAc
plt.plot(x, np.log10(total), "k-")

plt.ylim(-8.2,-0.8)  # Set y-axis range
plt.show()
```